K8s-08

1. **Create and test a Kubernetes Pod with an Empty Dir Volume**

- **Create an Empty Dir Volume yaml file**

```
root@Master:~/k8s-08# vi emptydir-pod.yaml
root@Master:~/k8s-08#
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: emptydir-pod

spec:
  containers:
  - name: nginx
    image: nginx

    volumeMounts:
    - name: cache-volume
      mountPath: /usr/share/nginx/html

  volumes:
  - name: cache-volume
    emptyDir: {}
```

- **Apply and verify it**

```
root@Master:~/k8s-08# kubectl apply -f emptydir-pod.yaml
pod/emptydir-pod created
root@Master:~/k8s-08# kubectl get pod emptydir-pod
NAME           READY   STATUS    RESTARTS   AGE
emptydir-pod   1/1     Running   0          6s
root@Master:~/k8s-08#
```

- **Create a file inside the volume and exit**

```
root@Master:~/k8s-08# kubectl exec -it emptydir-pod -- sh
# echo "Hello EmptyDir" > /usr/share/nginx/html/index.html

cat /usr/share/nginx/html/index.html# #
Hello EmptyDir
# exit
```

- **Now verify from outside it**

```
root@Master:~/k8s-08# kubectl exec emptydir-pod -- cat /usr/share/nginx/html/index.html
Hello EmptyDir
root@Master:~/k8s-08#
```

2. **Configure a Host Path Volume in Kubernetes and Validate Data Persistence**

- **Create a directory on the node and verify it**

```
root@Master:~/k8s-08# mkdir -p /data/hostpath
echo "HostPath Volume Test" > /data/hostpath/index.html
root@Master:~/k8s-08# cat /data/hostpath/index.html
HostPath Volume Test
root@Master:~/k8s-08#
```

- **Now Create host path-pod. Yaml**

```
root@Master:~/k8s-08# vi hostpath-pod.yaml
root@Master:~/k8s-08#
```

```
apiVersion: v1
kind: Pod
metadata:
  name: hostpath-pod

spec:
  containers:
  - name: nginx
    image: nginx

    volumeMounts:
    - name: host-volume
      mountPath: /usr/share/nginx/html

  volumes:
  - name: host-volume
    hostPath:
      path: /data/hostpath
      type: Directory
```

- **Apply the yaml file and verify pod**

```
root@Master:~/k8s-08# kubectl apply -f hostpath-pod.yaml
pod/hostpath-pod created
root@Master:~/k8s-08# kubectl get pod hostpath-pod
NAME           READY   STATUS    RESTARTS   AGE
hostpath-pod   1/1     Running   0          7s
root@Master:~/k8s-08#
```

- **Verify the mounted file**

```
root@Master:~/k8s-08# kubectl exec hostpath-pod -- cat /usr/share/nginx/html/index.html
HostPath Volume Test
root@Master:~/k8s-08#
```

- **Test data persistence (Create another file inside the pod) and exit**

```
root@Master:~/k8s-08# kubectl exec -it hostpath-pod -- sh
# echo "Persistent Data" > /usr/share/nginx/html/test.txt
# exit
root@Master:~/k8s-08#
```

- **Verify from the node**

```
root@Master:~/k8s-08# cat /data/hostpath/test.txt
Persistent Data
root@Master:~/k8s-08#
```

3. **Deploy an Amazon EBS Volume Using Persistent Volume and Persistent Volume Claim (PVC)**

- **Create pv.yaml file**

```
root@Master:~/k8s-08# mkdir -p /data/pv
root@Master:~/k8s-08# vi pv.yaml
root@Master:~/k8s-08#
```

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: host-pv
spec:
  capacity:
    storage: 2Gi
  accessModes:
    - ReadWriteOnce
  persistentVolumeReclaimPolicy: Retain
  hostPath:
    path: /data/pv
```

- **Apply and verify it**

```
root@Master:~/k8s-08# kubectl apply -f pv.yaml
persistentvolume/host-pv created
root@Master:~/k8s-08# kubectl get pv
NAME      CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS      CLAIM   STORAGECLASS   V
host-pv   2Gi        RWO            Retain           Available                          <
root@Master:~/k8s-08#
```

- **Create Pvc.yaml file**

```
root@Master:~/k8s-08#
root@Master:~/k8s-08# vi pvc.yaml
root@Master:~/k8s-08#
```

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: host-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
```

- **Apply and verify pvc and pv also**

```
root@Master:~/k8s-08# kubectl apply -f pvc.yaml
persistentvolumeclaim/host-pvc created
root@Master:~/k8s-08# kubectl get pvc
NAME       STATUS   VOLUME    CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
host-pvc   Bound    host-pv   2Gi        RWO                           <unset>                 7s
root@Master:~/k8s-08# kubectl get pv
NAME      CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAIM             STORAGECLASS   VOLUMEATTRIB
host-pv   2Gi        RWO            Retain           Bound    default/host-pvc                 <unset>
root@Master:~/k8s-08#
```

- **Create pv-pod.yaml file**

```
root@Master:~/k8s-08#
root@Master:~/k8s-08# vi pv-pod.yaml
root@Master:~/k8s-08#
```

```
apiVersion: v1
kind: Pod
metadata:
  name: pv-pod
spec:
  containers:
  - name: nginx
    image: nginx
    volumeMounts:
    - name: data
      mountPath: /usr/share/nginx/html

  volumes:
  - name: data
    persistentVolumeClaim:
      claimName: host-pvc
```

- **Apply and verify it**

```
root@Master:~/k8s-08# kubectl apply -f pv-pod.yaml
pod/pv-pod created
root@Master:~/k8s-08# kubectl get pod pv-pod
NAME     READY   STATUS    RESTARTS   AGE
pv-pod   1/1     Running   0          11s
root@Master:~/k8s-08#
```

- **Test the PV/PVC -** Create a file inside the pod:

```
root@Master:~/k8s-08# kubectl exec -it pv-pod -- sh
# echo "PV PVC Working" > /usr/share/nginx/html/test.txt
# exit
```

- **Now exit and check outside output**

```
root@Master:~/k8s-08# cat /data/pv/test.txt
PV PVC Working
root@Master:~/k8s-08#
```

4. **Set Up an Amazon EFS Volume and Attach it to Multiple Pods**

- **Create shared-pod1.yaml**

```
root@Master:~/k8s-08# mkdir -p /data/shared
root@Master:~/k8s-08# vi shared-pod1.yaml
root@Master:~/k8s-08#
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: shared-pod1
spec:
  containers:
  - name: nginx
    image: nginx
    volumeMounts:
    - name: shared-storage
      mountPath: /shared

  volumes:
  - name: shared-storage
    hostPath:
      path: /data/shared
      type: DirectoryOrCreate
```

- **Apply shared-pod1.yaml file**

```
root@Master:~/k8s-08# kubectl apply -f shared-pod1.yaml
pod/shared-pod1 created
root@Master:~/k8s-08#
```

- **Create shared-pod2.yaml file**

```
root@Master:~/k8s-08# vi shared-pod2.yaml
root@Master:~/k8s-08#
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: shared-pod2
spec:
  containers:
  - name: nginx
    image: nginx
    volumeMounts:
    - name: shared-storage
      mountPath: /shared

  volumes:
  - name: shared-storage
    hostPath:
      path: /data/shared
      type: DirectoryOrCreate
```

- **Apply shared-pod2.yaml file**

```
root@Master:~/k8s-08# kubectl apply -f shared-pod2.yaml
pod/shared-pod2 created
root@Master:~/k8s-08#
```

- **Verify both pods**

```
root@Master:~/k8s-08# kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
nginx-deployment-f9db776c5-qfcvj      1/1     Running   0          5m28s
recreate-deployment-7c6d446b89-tps2m  1/1     Running   0          5m28s
shared-pod1                           1/1     Running   0          3m22s
shared-pod2                           1/1     Running   0          54s
```

- **Write a file from Pod 1 and exit**

```
root@Master:~/k8s-08# kubectl exec -it shared-pod1 -- sh
# echo "Shared Storage Works" > /shared/demo.txt
# exit
```

- Read the same file from **Pod 2**:

```
root@Master:~/k8s-08# kubectl exec shared-pod2 -- cat /shared/demo.txt
Shared Storage Works
root@Master:~/k8s-08#
```

5. **Implement and Test Liveness and Readiness Probes in a Kubernetes Pod**

- **Create probe-pod. Yaml**

```
root@Master:~/k8s-08# vi probe-pod.yaml
root@Master:~/k8s-08#
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: probe-pod

spec:
  containers:
  - name: nginx
    image: nginx

    ports:
    - containerPort: 80

    livenessProbe:
      httpGet:
        path: /
        port: 80
      initialDelaySeconds: 10
      periodSeconds: 5

    readinessProbe:
      httpGet:
        path: /
        port: 80
      initialDelaySeconds: 5
      periodSeconds: 5
```

- **Apply and verify it**

```
root@Master:~/k8s-08# kubectl apply -f probe-pod.yaml
pod/probe-pod created
root@Master:~/k8s-08# kubectl get pod probe-pod
NAME        READY   STATUS    RESTARTS   AGE
probe-pod   1/1     Running   0          11s
root@Master:~/k8s-08#
```

- **Verify pods**

```
root@Master:~/k8s-08# kubectl describe pod probe-pod | grep -A10 "Liveness"
    Liveness:       http-get http://:80/ delay=10s timeout=1s period=5s #success=1 #failure=3
    Readiness:      http-get http://:80/ delay=5s timeout=1s period=5s #success=1 #failure=3
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-wk9gg (ro)
Conditions:
```

- **Simulate a liveness probe failure**
- **Delete the default NGINX page inside the container:**

```
root@Master:~/k8s-08# kubectl exec -it probe-pod -- rm /usr/share/nginx/html/index.html
root@Master:~/k8s-08# kubectl get pod probe-pod -w
NAME        READY   STATUS    RESTARTS      AGE
probe-pod   1/1     Running   0             2m18s
probe-pod   0/1     Running   0             2m20s
probe-pod   0/1     Running   1 (1s ago)    2m22s
probe-pod   1/1     Running   1 (9s ago)    2m30s
```

- **Now inspect it:**

```
root@Master:~/k8s-08# kubectl describe pod probe-pod
Name:             probe-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             master/172.31.37.70
Start Time:       Sun, 05 Jul 2026 07:42:44 +0000
Labels:           <none>
Annotations:      cni.projectcalico.org/containerID: 3249211f2a3907dea37e80adf8e86a26b3791af24af9
                  cni.projectcalico.org/podIP: 192.168.219.126/32
                  cni.projectcalico.org/podIPs: 192.168.219.126/32
Status:           Running
IP:               192.168.219.126
IPs:
Events:
  Type     Reason     Age                  From               Message
  ----     ------     ----                 ----               -------
  Normal   Scheduled  2m52s                default-scheduler  Successfully assigned default/probe-pod to mast
  Normal   Pulled     2m52s                kubelet            Successfully pulled image "nginx" in 141ms (141
  Warning  Unhealthy  33s (x3 over 43s)    kubelet            Liveness probe failed: HTTP probe failed with s
  Normal   Pulling    32s (x2 over 2m52s)  kubelet            Pulling image "nginx"
  Normal   Created    32s (x2 over 2m52s)  kubelet            Created container: nginx
  Normal   Started    32s (x2 over 2m52s)  kubelet            Started container nginx
  Warning  Unhealthy  32s (x5 over 43s)    kubelet            Readiness probe failed: HTTP probe failed with
  Normal   Killing    32s                  kubelet            Container nginx failed liveness probe, will be
  Normal   Pulled     32s                  kubelet            Successfully pulled image "nginx" in 136ms (136
```